Platform-:
Platforms /services where you can access OpenAI models
Platform.openai.com
https://ai.google.dev/

Platform.openai.com
Models:
GPT family models
Image generation models
Speech/audio models
Embedding models

https://ai.google.dev/

Models such as-:
Gemini family models
Embedding's models
Multimodal models

Platform can provide --> access to the models

Model --> generate the answer

Create an agent -

1. Generate an API key from https://ai.google.dev/
2. Create an environment where your project is going to create
3. Create folder :- name
   a. .env
   b. App.py
   c. Requerement.txt
4. Install packages, modules ..
   pip install openai ----> (Platform.openai.com)

pip install requests
pip install flask
pip install fastapi
pip install python-dotenv
pip install google-generativeai==0.5.2 -- SDK/package used to access google's gemini/generative AI models from the python code

Or pip install google-genai

pip show google-generativeai==0.5.2

Memory management :_
Memory management in agent creation means how AI agent stores, updates, retrieves, prioritize , and forgets information, so it can works efficiently across conversation or workflow ;-

Why memory matters in Agents:-

- As agent often need to remember:-
  - User preferences
  - Previous messages
  - Task progress
  - Tools already used
  - Error encountered
  - Environment details
  - Long term knowledge sources :
- Example -: user:-my kubernates cluster is prod-east
- Next time --- scale ngnix to 5 replicas
- Wirth memory , the agent know where agent apply the changes

Types of memory in Agent :-

1. Short term Memory (working memory)
   Used during current session / task
   Store:-

a. recent chat history
b. Current objective
c. Intermediate reasoning state
d. Last tool output
e. Example:- current incident :
   Pod failing
   Checked logs already

Next step:- inspects the events

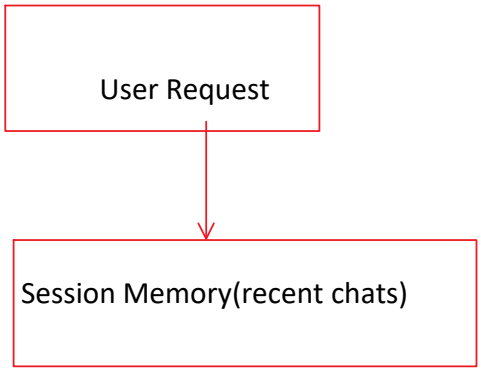
2. Long term memory:-
   Persists across session
   Store:-
   - Recurring preferences
   - Historical incidents
   - Prior successful fixes
   - User/team conventions
   - Examples - "use Helm, not raw YAML"
     "prod cluster name is aks-prod"
3. episodic memory :-
   Remember past events/task
   Example :- last month payemnt-service failed due to expired secret
4. Semantic memory :-
   i. General structured knowledge .

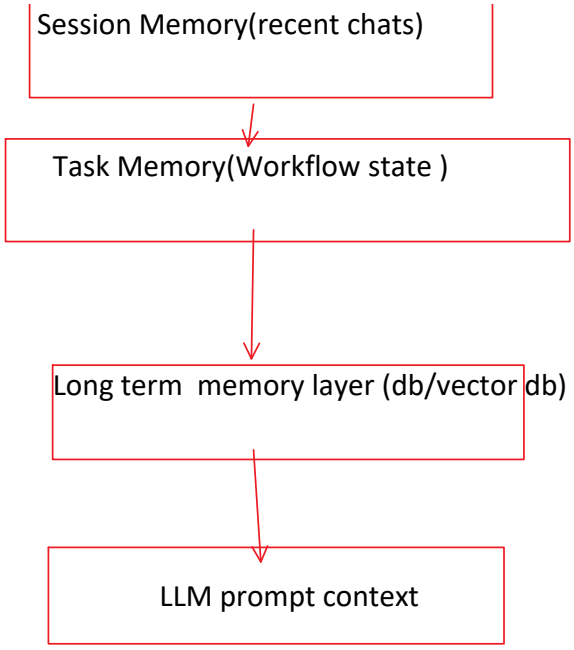
What memory management actually includes:-

A. Store :
   a. What the information should be saved?
   b. Not everything should kept
B. Retrieve
   a. When needed , fetch relevant memory
C. Update :
   a. Replace Outdated information (facts)
      Example-- old cluster :dev-01
      new cluster :prod-main
D. Forget/Expire :
   Remove stale or low-value memory .

E. Prioritize:
   a. Most relevant memory should be surfaced first

Memory layers in your agent:-





Technologies used :-

Simple storage :-

Python dict

Redis

sQlite

PostgreSql

Semantic retrieval :-

Vector DBs:

Chroma, pinecone,weaviate, FAISS

Example- memory={}

Memory["cluster"] = "prod-West"

Memory["NameSpace"] = "payments"

Print(memory["cluster"])

Redis -->

Import redis

r=redis.Redis(host="localhost",port=800)

r.set("cluster","prod-west")

Print(r.get(c"luster")

Interview Agent:-

Prompt :- act as a trainer, you have good knowledge and good skill son python language , can you please explain me the concept of OOPs in

python complete detailed information ,syntaxes, use cases and also give some case studies to understand and practice for the good